

TMMModel_Documentation

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Document Version: 1.2

Change History

Version	Changes
1.1	Added notes to 2.5 Destruction Models and 2.6 Animated Destruction Models to detail some FBX node naming and structure expectations.
1.2	Added notes related to changes from version 35 to 37 of the TMMModel format. • Added section 2.2.5 AutoAttachInfo Type. • Updated section 5.1 to state that the current version is 37 and that versions 35 onward can now load. • Added section 5.11 Auto Attach Properties to detail the extra properties that appear in version 37.

1 Introduction

The purpose of this document is to define information to help with understanding the Model System for TMMModels used by Age of Mythology: Retold.

Disclaimer: Although customization of model files is possible there may be untested situations where custom models may cause significant performance degradation or crashes.

2 FBXImport Files

2.1 Overview

FBX Import files define how a model or animation is turned into a TMM or TMA file type by providing contextual metadata.

2.2 Available Properties

Property	Type	Description
type	String	Defines the type of model and sets expectations for how the model can be used. Valid values include:

		• static • skeletal • destruction • animation • speedtree
adjustment_transform	Object	Defines an adjustment to the model. This is baked into the result when generating TMM/TMA files. Object contains: "t" as the translation component with x, y, z as a float array. "r" as the rotation component with x, y, z as a float array. "s" as the scale component with x, y, z as a float array.
skeleton	String	Path to the reference skeleton file. Used for skeletal animation and destruction with animation types.
include_vertex_colors	Boolean	Bakes the vertex colors from the model into the TMMModel while importing when defined. Destruction uses this typically for burnt interiors. Only enable if necessary.
IsTerrainEmbellishment	Boolean	This option is used for automatic LODing. Only intended to be used for bushes/rocks/etc.
EnableRayTracingForModel	Boolean	Enables this model to be processed by Ray Tracing.
auto_burn_mode	String	Valid values are: • NoAutoBurn = Does nothing. • AutoBurnClothDefault = Causes burning of cloth behaviors.
use_mikktspace	Boolean	Enabled by default. Recommended to keep enabled.
use_voxels_for_click_volume	String	Valid values are: • Auto = Uses voxels for selection if the model is static or generates selection spheres based on the bones in the model if animated.

		ForceVoxelData = Forces voxel mode. Useful for models that are animated with bones that don't define the real selection area properly. E.g., animated buildings and ships.
bone_click_volume_override	Object Array	Allows overriding of individual bone selection volumes when using the sphere selection mode. Each object in the array is defined with: - boneName = String that matches a bone in the model. - volumeMultiplier = Float that modifies the scale of the bone size to make the sphere larger or smaller.
Shared_Animation_Bucket_Count	Boolean	Defines the number of shared animation buckets to optimize shared re-use of animations (e.g., for trees) Defaults to 0. Note: For SpeedTree type assets, if this value is not set, it will default to 4.
boundingbox_padding	Float	Defaults to 0.0f. Adds padding to the bounding box of the model on every axis.
attachments	Object Array	See 2.2.1 for definition. Contains a list of attach points.
destructionoverrides	Object	See 2.2.2 for definition. Contains overrides for destruction defaults.
animation_controllers	Object Array	See 2.2.3 for definition. Contains a list of animation controllers.
physicsobjectsettings	Object	See 2.2.4 for definition. Contains definition of the physics properties for the model.
footprint_settings	Object	Deprecated property.

2.2.1 Attachments Object Type

Each attachment can have the following properties defined.

Property	Type	Description
dummy	String	Name of the attachment to act as a pseudo bone name.

bone	String	The name of the bone this object is parented to.
preview	String	Reference to a TMMModel relative path. Intended for showing previews of the model at the attach point in developer tools. Does nothing in game.
DummyBoneMode	String	Valid values: • "Auto" • "DoNotDummyBone" • "AddToSim" • "ForceToDummyBoneType" Information about modes: Controls whether the bone is added to the simulation. Auto - AddToSim if detected to be a dummy bone type (otherwise ignored). DoNotDummyBone - Ignores for Sim. AddToSim - Adds to the Sim (even if not paired as a dummy bone). ForceToDummyBoneType - Forces to a specified dummy bone type. Defaults to Auto if not specified.
DummyBoneTransformMode	String	• Auto (Auto chooses the most appropriate mode depending on what is available.) • BindPoseOnly (Captures only the bind pose and no animation data) • BindPoseAnimationFirstFrameOnly (Captures only the bind pose and first frame of an animation.) • BindPoseSpecificFrameOnly (Captures a bind pose from a specific animation frame only.) • AllFrames (Captures all frames of an animation.) • LimitedFrames (Captures all frames with a cap used to reduce the total count by only accepting some of the frames if over the minimum.) Defaults to Auto if not specified.

DummyFramePosition	Float	Defaults to 0 if not specified. Depends on the value of the Dummy Bone Transform Mode to determine if this is relevant. Used for any option where specified frames are referenced.
DummyFrameLimit	Integer	Defaults to -1 if not specified. Maximum number of frames to capture for sim version of bone path.
ForcedDummyBoneName	String	Value is only relevant if set to Dummy Bone Mode ForceToDummyBoneType. Changes the name to be treated as this for attach points in simulation. Defaults to empty if not specified.
DummyBoneAnimationFilter	String	• Auto (Auto chooses depending on what is available) • AllAnimations (Considers all animations that are detected to be played against the model) • OnlySingleAnimation (Similar to OnlySpecificAnimations except only allows one and can allow for optimizations with some configurations.) • OnlySpecificAnimations (Only considers the specific listed animations) • AllExceptSpecificAnimations (Considers all detected animations except for the listed items.)
DummySpecificAnims	String Array	A list of relative references to animations. The behavior depends on what the DummyBoneAnimationFilter is set to.

2.2.2 DestructionOverrides Type

Each object in the DestructionOverrides object array expects a single one of the following properties. The existence of these properties in order from top to bottom will halt loading when it finds one to use as that type.

Member Name	Description
globaloverrideType	Defines an override for a specified property.
perproxyoverride	Defines an override for a specific proxy to change the force direction.

impactoverride	Modifies an import point for a specific proxy.
usecustominter valdata	Defines custom intervals for how proxy groups should break at different health thresholds.

If none of these are specified the expectation is that “scope”, “property” and “value” fields are specified for a defined override. (See scoped override).

2.2.2.1 Global Override Types

Each object of type globaloverridetype is defined by these properties:

Property	Type	Description
globaloverridetype	String	The type to override/
value	String	Value to override to.

Override types include the following:

GlobalOverrideType	Description
VFXDebrisOnDeath	Relative path to a Composite Model that spawns on death instead of any default debris.
FullDestroyMode	Valid values include: • "All At Once" • "Top To Bottom" • "Bottom To Top" • "Outward From Point" • "Inward To Point" • "Random" • "Ship Default" • "Ship Kraken Animation" • "Not Set (Default Top To Bottom)"
VFXDebrisMode	Valid values include: • "Default (auto evaluated)" • "Building" • "WallConnector" • "WallShort" • "WallMedium" • "WallLong" • "WallGate" • "Ship"

	• "Land Unit" • "No Debris"
VFXDebrisCivType	Valid Values include: • "Greek" • "Egyptian" • "Norse"

2.2.2.2 Per Proxy Override Types

Each object of type perproxyoverride is defined by these properties:

Property	Type	Description
perproxyoverride	Integer	Defines the proxy ID that this override is modifying.
forcedirection	Float Array	x, y, z as 3 Floats forming a direction vector.

2.2.2.3 Impact Override Types

Each object of type impactoverride is defined by these properties:

Property	Type	Description
impactoverride	Integer	Defines the proxy ID that this override is modifying.
impactposition	Float Array	x, y, z as 3 Floats forming a local proxy space position for impacting.

2.2.2.4 Use Custom Intervals Override Types

Each object of type usecustomintervaldata is defined by these properties:

Property	Type	Description
usecustomintervaldata	Boolean	Toggles whether this content is used.
intervals	Object Array	Array of interval definitions.

Each interval object consists of the following:

Property	Type	Description
eventthreshold	Float	Health percent to use for triggering when the value drops below this number.

groups	Integer Array	Array of Proxy Group IDs to trigger when the event occurs.
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2.2.2.5 Scoped Override

Scoped overrides expect the following properties to be defined.

Property	Type	Description
scope	String	Valid options include: “Global” (applies to everything) “Group” (applies to proxy group) “Chunk” (applies to specific chunk ID)
groupid	Integer	Defines proxy group ID when using the “Group” scope.
chunkid	Integer	Defines chunk ID when using the “Chunk” scope.
property	String	See 2.2.2.6 for all the valid values.
value	Varies	Depending on the property type this changes between Boolean, Integer, and Float types as expected values.

2.2.2.6 Valid Property Override Values

Note that values are floats unless otherwise shown as a “true” or “false” for the default or an integer form number for Integer.

Property	Default	Min	Max	Description
JittersForProxy	2	-1	-	Integer type. Min -1 = Never Jitter Ever, 0 = Jitter But Never from Hits, 1+ = That many jitters then destroyed. On Death only supports -1 or 0. Not usable in “Chunk” scope.
JitterIntensity	1.0	-	-	The intensity applied as a multiplier to scale the jitter effect. Proxy or Global ONLY!.
PercentToLeave	0.5	0.0	1.0	A value between 0 and 1 indicating a percent of hits to not apply before death.

				Only allowed to apply to Global Scope.
PhysicsDensityStone	3000.0	200.0	-	Overrides the default density applied per-physics object. Only applies to the specified type of physics object.
PhysicsDensityWood	720.0	200.0	-	Overrides the default density applied per-physics object. Only applies to the specified type of physics object.
PhysicsDensityMetal	7000.0	200.0	-	Overrides the default density applied per-physics object. Only applies to the specified type of physics object.
PhysicsDensityCloth	200.0	200.0	-	Overrides the default density applied per-physics object. Only applies to the specified type of physics object.
PhysicsRestitutionStone	0.2	0.0	-	Overrides the default resitution applied per-physics object. Only applies to the specified type of physics object.
PhysicsRestitutionWood	0.3	0.0	-	Overrides the default resitution applied per-physics object. Only applies to the specified type of physics object.
PhysicsRestitutionMetal	0.7	0.0	-	Overrides the default resitution applied per-physics object. Only applies to the specified type of physics object.
PhysicsRestitutionCloth	0.0	0.0	-	Overrides the default resitution applied per-physics object. Only applies to the specified type of physics object.
PhysicsPenetrationDepth	1e-4f	1e-4f	-	Penetration depth controls how much the physics object can overlap with other object.

IsPhysicsControlled	true	-	-	Defaults to true. Set this to false to force objects to instantly disappear when they break off.
MotionStopHideDelay	1.0	0.0	-	Delay before starting to count down and remove. Total is sum of this with the stop hide time and a random delay extension. Not usable in “Chunk” scope.
MotionStopHideTime	4.0	0.0	-	Duration of fade out. Total is sum of this with the stop hide delay and a random delay extension. Not usable in “Chunk” scope.
MotionStopHideDelayRandomExtension	0.5	0.0	-	Random 0 to N time in seconds variation on chunks disappearing. Not usable in “Chunk” scope.
ForceMultiplier	1.0	-	-	Multiplies the force applied to all chunks that are part of the affected group. Not usable in “Chunk” scope.
KeepHoudiniTargetTypes	false	-	-	Enables use of target types set in Houdini. This is set to false by default. When false, this property will force ignore height settings from Houdini and automatically generate them. Only usable in “Global” scope.
EnableAngularVelocity	true	-	-	Enables or disables angular velocity.
EnableProxyGroupShapes	true	-	-	Enables/Disables spawning of all proxy group shapes. Disabling this will result in objects likely falling straight down. Only available as a global

				option that applies to all proxy shapes on this object.
VFXAllowDecal	true	-	-	Enables/Disables decals for a proxy or globally.
VFXAllowPopcorn	true	-	-	Enables/Disables popcorn for a proxy or globally.

2.2.3 AnimationControllers Type

Each of the following properties are included under the “attachments” array.

Property	Type	Description
type	String	Valid options: • attach_point_visibility • attach_point_footprint This property is required.

Attach Point Visibility Properties:

These additional properties are necessary if the type is set to attach_point_visibility. The purpose of this animation controller is to hide attachments during specific periods of animation. E.g. weapon is put away to do something.

Property	Type	Description
start_time	Float	Start time of the event as a percent of the animation time (expects 0 to 1 value).
end_time	Float	End time of the event as a percent of the animation time (expects 0 to 1 value).
ease_in_time	Float	Time before the event time to transition.
ease_out_time	Float	Time after the event time to transition.
invert_logic	Boolean	Inverts the visibility state to the opposite of normal behavior.
attachpoint	String	Name of the attach point defined in the separate “attachments” array.

Attach Point Footprint Properties:

These additional properties are necessary if the type is set to attach_point_footprint. The purpose of this controller is to spawn a temporary decal that remains after the unit is moving (typically walk animations).

Property	Type	Description
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spawn_time	Float	Time for the event to trigger.
footprint_name	String	Name of the footprint to use.
footprint_id	Integer	Index of the footprint to use.
invert_texture_y	Boolean	Inverts the texture on the Y axis when enabled.
is_right_side	Boolean	Sets whether it is treated as the right side (or left side). When enabled this will flip the X axis of the footprint texture.
attachpoint	String	Name of the attach point defined in the "attachments" array to link against.

2.2.4 PhysicsObjectSettings Type

Physics object shape settings that allow mostly for things like barrels or trees to fall with physics settings. Expects the following properties.

Property	Type	Description
physics_enableshapeproperties	Boolean	Controls whether these settings do anything. Allows the properties to exist in the file but be disabled.
physics_shape_type	String	This controls what type of shape is generated. Valid options include: • "Box" • "Sphere" • "ConvexHull" • "NoShape" • "Unknown"
physics_effect_type	String	Valid options include: • "Cloth" • "Stone" • "Metal" • "Wood" • "Unknown"
physics_maxvertices	Integer	Cap on the vert count when using Convex Hull mode.
physics_isphysicscontr	Boolean	Generally on. This determines if the object freely moves as a rigid body when enabled.

olled		
physics_isfixedmotion	Boolean	Determines if the object uses fixed motion or Inertia.
physics_restitution	Float	Minimum of 0.0f. Typical values are roughly: • Stone = 0.2 • Wood = 0.3 • Metal = 0.7 • Cloth = 0.0
physics_density	Float	Typical values are roughly: • Stone = 3000.0 • Wood = 720.0 • Metal = 7000.0 • Cloth = 200.0
physics_penetrationdepth	Float	Depth object can intersect with other objects for collisions. Defaults to 1e-4f.

2.2.5 AutoAttachInfo Type

The following properties are related to automatic additional actions that can occur while the model is converted.

Property	Type	Description
autoAttachCorpseToBone	Bool	This controls where the skeleton will spawn with automatic placement. Defaults to true when not set.
deathAnimation	String	Forces the death anim to use the specified anim instead of any detected death anim. Tries to default detection of the death animation. This property defaults to empty if not set.
corpseAttachBoneName	String	Sets the bone where the skeleton will spawn. Auto will try and use the spine bone if one exists. Defaults to empty and will try to smart select a suitable bone or falls back to unit origin.

autoAddImpactPointsUsingIdle	Bool	Uses idle (or other specified) animation to generate impact points. Only intended for things with no destruction. Defaults to false. Turning this to true won't do anything if there is a physics object defined.
impactPointOverrideAnimIdle	String	Sets an override to specify the idle animation used for impacts. Defaults to empty. Will automatically try and look up an idle animation when not set.
manualimpactpoints	Object Array	Contains an array of objects with property: • “transform” with a 3 float vector representing a translation. These will be added automatically as additional ranged attack points and controls the position of spawned VFX from ranged attacks.

2.3 Static Models

Static models don't have a skeleton and are considered a simple mesh.

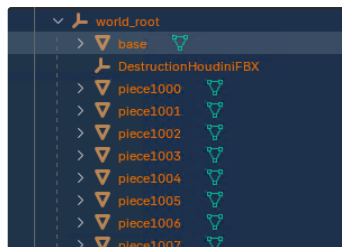
2.4 Animated Models

Animated models have a skeleton that can mutate based on a TMA file animation.

2.5 Destruction Models

Destruction type models expect both an fbximport file and a destructionimport file. This refers to non-animated destruction models. Destruction enabled models auto define a flat heirarchy skeleton to reference all the destruction chunk objects. These are static models with the addition of destruction information.

Destruction FBX files expect a certain node structure.

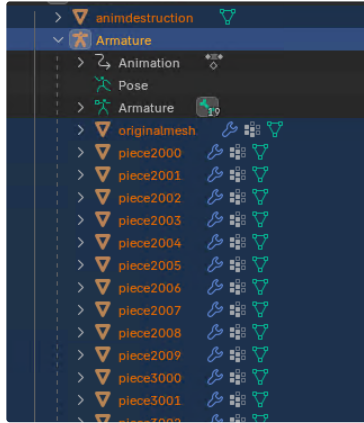


- This data should match the information stored in your destructionimport data otherwise data may be ignored during import or behave incorrectly.
- The parent is expected to be called “world_root”.
- Up to a single node can be called “base” and this should contain the data for your base. (optional)
- A single node is expected called “DestructionHoudiniFBX” with no content to flag that the model is meant to be interpreted as destruction. (This is different for animated destruction).
- Any number of “pieceXXXX” nodes that define each chunk of the destruction mesh. The XXXX should be unique.
- Nodes are looked up by name to match against chunk definitions so the node naming is important.

2.6 Animated Destruction Models

Destruction models combined with a skeleton so that they have two skeletons. This does incur more of a cost so it is recommended to limit these more than regular destruction models if possible. When chunks break off they swap from animating using the bone weights for the chunk to using the destruction skeleton.

Animated destruction models expect a specific node structure.



- animdestruction must exist as a top level node to flag that this model is intended to behave as animated destruction.
- An Armature node is expected with children for all other nodes.
- The originalmesh node contains the undamaged original version of the mesh.
- The simplemesh node contains a simplified version of the original mesh.
- The remainder are expected to be “pieceXXXX” nodes similar to regular destruction.

2.7 Speed Tree Models

The following properties are only relevant if the model is a speed tree. These settings are contained under a “speedtree” JSON object.

Property	Type	Description
do_chop	Boolean	Only relevant if model “type” is set to SpeedTree. When enabled, this will enable chop behavior that cuts the model into multiple sub-models. One with the tree whole, one with the tree cut into two pieces with bones for destruction automatically added, and one with only the stump. The following properties control the chop.
chop_height	Float	Only relevant if model “type” is set to SpeedTree when do_chop is enabled. Sets the height of where the chop will happen. Typical values between 0 and 1.

chop_slice_size	Float	Only relevant if model “type” is set to SpeedTree when do_chop is enabled. The radius of the cut area. It should be larger than the size. Typical values between 0 and 0.5.
chop_extrude_point	Float	Only relevant if model “type” is set to SpeedTree when do_chop is enabled. Controls the distance away from the chop height to clear away in cut versions. It should be larger than the size. Typical values between 0 and 0.5.
chop_physics_pivot_height	Float	Only relevant if model “type” is set to SpeedTree when do_chop is enabled and the simple physics object is defined. Controls the absolute Y position above the terrain used for a cone peak at the base of the physics object. This can be used to tweak how pointy the physics object is and how far it “falls” when swapping to a chopped version. Values are typically between 0 and 1.
chop_physics_clamp_xz	Float	Only relevant if model “type” is set to SpeedTree when do_chop is enabled and the simple physics object is defined. 0.0 is no clamp. Clamping will shrink in any points relative to the model center as a radius around that point. Only points above the threshold will be clamped. This can be used to make trees roll or not roll a bit better while falling. Typical values range between 0 and 10.

Trees should normally default to wood type physics object settings, with convex hull mode set to max of 20 verts.

2.8 Animation Type Models

The following properties are only relevant if the model is an animation type. Animation type models turn into TMA files.

Property	Type	Description
resample_animation_frames	Integer	Resample Frame Count. The alternate number of frames to resample as during export.
resample_animation_looped	Boolean	Resamples as a looped animation.

override_animation_length	Float	Override animation length (in seconds).
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3 DestructionImport Files

These files define additional information used for destruction. As part of internal tools this is generated in Houdini.

*.destructionimport files are JSON.

Property	Type	Description
HoudiniTool Version	Integer	Expects the number value to be 3.
isAnimated Destruction	Boolean	
hasBurntInterior	Boolean	
sha256FBX Hash	String	Required field. Hash of the source FBX file this import data is paired with. This is computed in python using: <pre> 1 import hashlib 2 3 sha256HashSimple = hashlib.sha256() 4 5 with open(simpleFilePath, "rb") as f: 6 # Update hash string value in blocks 7 # of 4K 8 for readBytes in iter(lambda: 9 f.read(4096), b''): 10 sha256HashSimple.update(readBytes) 11 12 hashResultSimple = 13 sha256HashSimple.hexdigest() </pre>
proxyGroups	Object Array	Required field. Holds a list of proxy group definitions. Defined in section 3.1.
leftoverChunks	Object Array	Required field. Holds a list of chunks that are not part of proxy groups. These objects are defined in section 3.2.

3.1 Proxy Group Objects

Each proxy group object is defined with the following properties. All fields are considered required fields. Proxy groups represent a collection of chunks that are close together and will explode out at the same time.

Property	Type	Description
proxyGroup ID	Integer	Unique index of the proxy group.

point	Float Array	X, Y, Z as 3 floats.
isLow	Boolean	Does nothing.
isTitan	Boolean	Does nothing.
isHigh	Boolean	Does nothing.
proxyOrder	Integer Array	Allow with a list of proxyGroupIDs for other proxy groups that must be destroyed as dependencies for this group to be destroyed.
chunks	Object Array	Array of Chunk objects matching the definition in section 3.2.

3.2 Chunk Objects

Note that each chunk object reference should be unique and not paired with more than one proxy group or leftover chunks. All properties are considered required fields.

Property	Type	Description
chunkName	String	Name matching the mesh name of the associated chunk.
proxyGroup ID	Integer	Paired proxy group ID or -1 if part of leftover chunks.
isBase	Boolean	Flags whether this is considered the base chunk that never collapses. There is only really expected to be a single base chunk.
boundingBoxSize	Float Array	X, Y, Z as a 3 floats.
boundingBoxTranslation	Float Array	X, Y, Z as a 3 floats.
boundingBoxRotation	Float Array	X, Y, Z as a 3 floats.
boundingBoxScale	Float Array	X, Y, Z as a 3 floats.
destructionPhysicsType	String	Expects one of the following: • wood • metal • cloth • stone • unknown

4 Material Files

Material files define material information for how a TMMModel is rendered. One is expected that matches the name of the TMM file. Material files are XML format.

4.1 Parent Node Attributes

These attributes can be set on the root node.

Attribute	Type	Description
bypassNameCheck	Boolean	Normally should not be enabled. This bypasses the normal validation checks that expect materials to be called MatA ... MatZ type format. Recommended to use these names.

4.2 Submaterial Nodes

Each child node of the parent are expected to be <submaterial> type nodes.

Node Type	Description
materialdef	Only one of this node is expected. Expects an attribute “name” to be set to the material type.
parameters	Contains a list of parameters that are relevant to the specified materialdef. The first parameters group should have no attributes. Any additional groups of parameters should have a “variant” attribute set to the index number of the variant. This is used to change which material variant is shown by the model instance. To view the full list of parameters and data related to each materialdef you can look inside the matching named file Render.bar under game/Render/.
permutations	Each group of permutations is always expected to specify the variant number as a “variant” attribute. Consists of a group of <permutation> nodes each with a specified valid permutation name as the value.

5 TMM File Definition

The following sections define the breakdown of all the content inside a TMM file.

5.1 Initial Header

Name	Type	Info
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Header	4 bytes (4 chars)	Value is always "BTMM"
Version	4 bytes DWORD	Currently, version 37 is the latest and expected. Models from version 35 onward are currently allowed to load, but may be missing features.
Bounds Min	16 bytes (Vector 4 floats)	Tight bounds
Bounds Max	16 bytes (Vector 4 floats)	Tight bounds
True Bounds	16 bytes (Vector 4 floats)	True bounds (used for visual culling - should be large enough to include max extended bones)
True Bounds	16 bytes (Vector 4 floats)	True bounds (used for visual culling - should be large enough to include max extended bones)
Bounds Radius	4 bytes float	
Primitive Count	4 bytes DWORD	
Material Count	4 bytes DWORD	
Submodel Count	4 bytes DWORD	
Bone Count	4 bytes DWORD	
Shared Animation Bucket Count	4 bytes DWORD	Used for sharing vertex animation on trees (typically something like 2 to 5 when used).
Attachment Count	4 bytes DWORD	
Vertex Buffer Size	4 bytes DWORD	Should match count in TMM.data file
Index Buffer Size	4 bytes DWORD	Should match count in TMM.data file
Vertex Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file

Vertex Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Index Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Index Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Skinning Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Skinning Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Destruction Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Destruction Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Color Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Color Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Local Height Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Local Height Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
Speed Tree Buffer File Offset	4 bytes DWORD	Should match count in TMM.data file
Speed Tree Buffer Size in Bytes	4 bytes DWORD	Should match count in TMM.data file
IsTerrainEmbellishment	1 byte (bool 0 or 1)	Used for automatic LOD of tree assets.
EnableRayTracingForModel	1 byte (bool 0 or 1)	
AdjustmentTransform	48 bytes (3x4 Matrix)	Does nothing during gameplay. This is used for tracking what adjustment transform was baked into the data buffers.

5.2 Attachments

The following properties are repeated, matching the number of attachments specified in the header as “Attachment Count”.

Name	Type	Info
Attach Type	4 bytes (Integer)	• DefaultBoneAttached = 0 (Attached to a bone on the skeleton (if bone index is a valid index). • TreeTrunk = 1 (Attached to the trunk of a tree that could be broken off due to destruction.) Note this is only expected if the dummy bone name is set to “SpecialAttachTreeTrunk”.
Bone Index	4 bytes (Integer)	
Dummy Bone Name	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Adjustment Transform Mutated Attach Matrix	48 bytes (3x4 Matrix)	Result of multiplying the local transform of the attachment point by the adjustment transform.
Local Transform	48 bytes (3x4 Matrix)	
Dummy Bone Mode	4 bytes (Integer)	• Auto = 0 (Auto choose correct dummy type if there is one or otherwise don't include for sim.) • DoNotDummyBone = 1 (Prevents use of this bone entirely for export to the sim database.) • AddToSim = 2 (Forces bone to be visible to the simulation (automatically becomes a dummy bone matched type if it is one). • ForceToDummyBoneType = 3 (Force bone to be visible to the sim and treat as a specific dummy bone type.)
Dummy Bone Transform Mode	4 bytes (Integer)	• Auto = 0 (Auto chooses the most appropriate mode depending on what is available.) • BindPoseOnly = 1 (Captures only the bind pose and no animation data) • BindPoseAnimationFirstFrameOnly = 2 (Captures only the bind pose and first frame of an animation.) • BindPoseSpecificFrameOnly = 3 (Captures a bind pose from a specific animation frame only.)

		• AllFrames = 4 (Captures all frames of an animation.) • LimitedFrames = 5 (Captures all frames with a cap used to reduce the total count by only accepting some of the frames if over the minimum.)
Force Dummy Bone Type String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	Value is only relevant if set to Dummy Bone Mode ForceToDummyBoneType.
Frame Limit	4 bytes (Integer)	Depends on the value of the Dummy Bone Transform Mode to determine if this is relevant.
Frame Position	4 bytes (Float)	Depends on the value of the Dummy Bone Transform Mode to determine if this is relevant.
Dummy Bone Animation Filter	4 bytes (Integer)	• Auto = 0 (Auto chooses depending on what is available) • AllAnimations = 1 (Considers all animations that are detected to be played against the model) • OnlySingleAnimation = 2 (Similar to OnlySpecificAnimations except only allows one and can allow for optimizations with some configurations.) • OnlySpecificAnimations = 3 (Only considers the specific listed animations) • AllExceptSpecificAnimations = 4 (Considers all detected animations except for the listed items.)
Dummy Specific Animations Count	4 bytes (Integer)	
For Each Index of "Dummy Specific Animations Count"	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	1 String per Count of Animations. These should be relative paths to animations that pair with the use of the Dummy Bone Animation Filter Setting when relevant.

5.3 Primitives

The following is repeated for the count listed in the header as "Primitive Count".

Name	Type	Info
Total Vertex Count for	4 bytes DWORD	

Primitive		
Total Index Count for Primitive	4 bytes DWORD	
Total Vertices Count for Primitive	4 bytes DWORD	
Total Faces for Primitive	4 bytes DWORD	
Material Index	4 bytes DWORD	The material this primitive uses.
Submodel Mask	4 bytes DWORD	Used for filtering out submodels for some situations (e.g., Trees that can be chopped have multiple states by masking).

5.4 Materials

The following is repeated for the count listed in the header as “Material Count”.

Name	Type	Info
Material Name String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	

5.4 Submodels

The following is repeated for the count listed in the header as “Submodel Count”.

Name	Type	Info
Submodel Name String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	

5.4 Bones

The following is repeated for the count listed in the header as “Bone Count”.

Name	Type	Info
------	------	------

Bone Name String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Bone Parent Index	4 bytes (Integer)	-1 = no parent
Bounds Center	16 bytes (Vector 4 floats)	Center of bone. Used for selection when not using voxel mode.
Bounds Radius	4 bytes (Float)	Size of sphere for selection. Used for selection when not using voxel mode.
Local Transform	64 bytes D3DXMATRIX	
Bind Pose	64 bytes D3DXMATRIX	
Inverse Bind Pose	64 bytes D3DXMATRIX	

5.5 Modified Bones

The following is ONLY written if “Bone Count” is greater than 0. Modified bones are used to mutate the radius of bones for selection. In some cases, the selection spheres are way too large or too small, causing strange selection behaviors. In these cases, the bones are modified.

Name	Type	Info
Count of Modified Bones	4 bytes DWORD	

The following is repeated for the count listed in “Count of Modified Bones”.

Name	Type	Info
Bone Index	4 bytes (Integer)	The bone this is modifying.
Bounds Radius Original	4 bytes (Float)	The radius value is kept here for simpler loading into the tool. This is a duplicate of what is stored in the bone value.
Bounds Radius Multiplier	4 bytes (Float)	The multiplier to modify the radius.

5.6 Auto Burn Mode

Name	Type	Info
Auto Burn Mode	1 BYTE	- NoAutoBurn = 0 - AutoBurnClothDefault = 1 Intended to be 0 unless the object has cloth, and you want it to apply burning based on unit health automatically.

5.7 Destruction

Destruction content is only relevant if the model has a destruction import file associated with it.

Name	Type	Info
Is Destruction Present	1 byte (bool 0 or 1)	If this value is false (0), skip all remaining parts of destruction.
Destruction Errors	4 bytes DWORD	Flag variable that should be 0 if there are no problems. The error values are listed below. <pre> 1 Default = 0, 2 BurntInteriorForceRemoved = (1 << 0), 3 CustomIntervalGroupMismatch = (1 << 1), 4 ScaleForceRemoved = (1 << 2), </pre>
Destruction Per Chunk Bones Count	4 bytes DWORD	The count of bones used for destruction.

The following is repeated for the count listed in “Destruction Per Chunk Bones Count”.

Name	Type	Info
Bone Name String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Bine Pose	64 bytes D3DXMATRIX	
Inverse Bind Pose	64 bytes D3DXMATRIX	

Continued general properties for destruction.

Name	Type	Info
Has Base	1 byte (bool 0 or 1)	A base is used to define a part of the model that will not collapse when an object is destroyed, e.g., the floor.

Base Chunk Index	4 bytes DWORD	Defaults to 0 if Has Base is false.
Enable Spawning of Proxy Group Shapes	1 byte (bool 0 or 1)	Controls whether physics shapes are spawned for the proxy groups.
Jitter Count On Death	4 bytes (Integer)	Part of the On Death group shared definition. (These are defined separately to the per group part because they have slightly less data). Should only be -1 or 0 because the on-death group can never jitter.
Jitter Intensity On Death	4 bytes (Float)	Part of the On Death group shared definition.
Motion Stop Hide Delay On Death	4 bytes (Float)	Part of the On Death group shared definition.
Motion Stop Hide Time On Death	4 bytes (Float)	Part of the On Death group shared definition.
Motion Stop Hide Delay Random Extension On Death	4 bytes (Float)	Part of the On Death group shared definition.
Force Multiplier On Death Group	4 bytes (Float)	Part of the On Death group shared definition.
Physics Type On Death Group	4 bytes (Integer)	Part of the On Death group shared definition. CUSTOM = 0 STONE = 1 WOOD = 2 METAL = 3 CLOTH = 4
Allow Decals VFX On Death	1 byte (bool 0 or 1)	Part of the On Death group shared definition.
Allow Popcorn VFX On Death	1 byte (bool 0 or 1)	
Percent To Leave	4 bytes (Float)	
Has Custom Debris	1 byte (bool 0 or 1)	

Custom Debris String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	ONLY present if Has Custom Debris is true (1). This should be a relative path to a composite model file.
Debris Type	4 bytes (Integer)	<pre> 1 // Auto evaluated 2 Default = 0, 3 // Do not spawn debris (will still do so if there is a custom override set) 4 NoDebris = 1, 5 // Buildings consider the footprint and civ when choosing types. 6 // Greek and Atlantean share the same effects. 7 // Norse and Chinese share the same effects. 8 // Egyptian has their own effects. 9 // Everything else defaults to greek. 10 Building = 2, 11 // Variation of building for only walls (still follows the same rules for civs as buildings) 12 BuildingWallConnector = 3, 13 // Variation of building for only walls (still follows the same rules for civs as buildings) 14 BuildingWallShort = 4, 15 // Variation of building for only walls (still follows the same rules for civs as buildings) 16 BuildingWallMedium = 5, 17 // Variation of building for only walls (still follows the same rules for civs as buildings) 18 BuildingWallLong = 6, 19 // Variation of building for only walls (still follows the same rules for civs as buildings) 20 BuildingWallGate = 7, 21 // Ships consider the size only 22 Ship = 8, 23 // Other units all just fall into a basic 1x1 for land units. 24 LandUnit = 9,</pre>
Debris Civ Type	4 bytes (Integer)	<pre> 1 Greek = 0, 2 Egyptian = 1, 3 Norse = 2,</pre>
Default Full Destroy Mode	4 bytes (Integer)	The mode used for collapsing during full destruction. <pre> 1 // Default state. MUST be changed before trying to defer. 2 DefaultNotSet = 0, 3 // Defer amount is defaulted to 0.0f for every chunk. 4 AllAtOnce = 1, 5 // Defer amount is set relative to the Y value of the chunk. 6 TopToBottom = 2, 7 // Defer amount is set relative to the opposite of top to bottom. 8 BottomToTop = 3, 9 // Defer amount is set relative to the reference point, with chunks closer to the point going first. 10 OutwardFromPoint = 4, 11 // Defer amount is set relative to the reference point, with chunks further away from the point going first. 12 InwardToPoint = 5, 13 // Random destroy mode between 1 and this enum -1. 14 Random = 6, 15 // Ship Default Destroy Mode (throws out parts a bit differently compared to the other destroy modes). 16 ShipDefault = 7, 17 // Kraken ship destruction staggers and applies forces matching the kraken animation. 18 ShipKrakenAnimation = 8,</pre>
Count of Proxy Groups	4 bytes DWORD	

The following is repeated for the count of the value “Count of Proxy Groups”.

Name	Type	Info
Proxy Center	16 bytes (Vector 4 floats)	
Impact Point	16 bytes (Vector 4 floats)	Position where impacts should appear to hit.
First Chunk Bone Index	4 bytes DWORD	Groups of bones are listed such that they appear as groups from this position in the array for the next element's count.
Chunk Count	4 bytes DWORD	
Jitter Count	4 bytes (Integer)	
Jitter Intensity	4 bytes (Float)	
Motion Stop Hide Delay	4 bytes (Float)	
Motion Stop Hide Time	4 bytes (Float)	
Motion Stop Hide Delay Random Extension	4 bytes (Float)	
Force Multiplier	4 bytes (Float)	Allows for extra force to help chunks overcome awkward surrounding physics shapes.
Force Direction Vector Default Value	16 bytes (Vector 4 floats)	The default direction vector is used if a custom value has not been set.
Force Direction Vector Real Value	16 bytes (Vector 4 floats)	If a custom vector has been set, this is what would be used. Otherwise, it should match the same as the default value.
Bounds Min	16 bytes (Vector 4 floats)	
Bounds Max	16 bytes (Vector 4 floats)	
Bounds Center	16 bytes (Vector 4	

	floats)	
Bounds Size	16 bytes (Vector 4 floats)	
Physics Type	4 bytes (Integer)	CUSTOM = 0 STONE = 1 WOOD = 2 METAL = 3 CLOTH = 4
Allow Decals VFX	1 byte (bool 0 or 1)	
Allow Popcorn VFX	1 byte (bool 0 or 1)	
Proxy Group Order Count	4 bytes DWORD	The count of how many dependent proxy groups there are that must be destroyed before this proxy group is allowed to be destroyed.

The following is repeated for the count defined in “Proxy Group Order Count”.

Name	Type	Info
Proxy Group Index	4 bytes DWORD	

Continued Per-Proxy Group data.

Name	Type	Info
Proxy Group Flags	4 bytes (Integer)	Currently unused.

Content after proxy group chunk for the count of proxy groups starts here.

Destruction Intervals Definition

Name	Type	Info
Count of Intervals	4 bytes (Integer)	

The following is repeated for the count defined in “Count of Intervals”.

Name	Type	Info
Event Threshold	4 bytes (Float)	The health threshold to trigger at.

Count of Chunk Groups To Pop At Interval	4 bytes (Integer)	
--	-------------------	--

The following is repeated for the count defined in “Count of Chunk Groups To Pop At Interval”.

Name	Type	Info
Proxy Index	4 bytes (Integer)	

5.8 Physics Template

Name	Type	Info
Has Physics Template	1 byte (bool 0 or 1)	If this is false, skip to after this section.
Physics Shape Type	4 bytes (Integer)	
Max Vertices	4 bytes (Integer)	
Effect Type for Export	4 bytes (Integer)	
Is Fixed Motion	1 byte (bool 0 or 1)	
Is Physics Controlled	1 byte (bool 0 or 1)	
Restituion	4 bytes (Float)	
Density	4 bytes (Float)	
Penetration Depth	4 bytes (Float)	
Count of Simple Physics Object Positions	4 bytes (Integer)	Point cloud used to create a convex hull.

The following is repeated for the count defined in “Count of Simple Physics Object Positions”.

Name	Type	Info
Local Space Position	16 bytes (Vector 4)	

	floats)	
--	---------	--

5.9 Tree Destruction Skeleton

Name	Type	Info
Has Tree Destruction Skeleton	1 byte (bool 0 or 1)	If this is false, skip to after this section.
Count of Generated Bones	4 bytes DWORD	

The following is repeated for the count defined in “Count of Generated Bones”.

Name	Type	Info
Bone Name String	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Bone Pose	64 bytes D3DXMATRIX	
Inverse Bind Pose	64 bytes D3DXMATRIX	

5.10 Click Volume

Name	Type	Info
Click Volume Type	1 BYTE	Auto = 0, // Use Skeletal pose if available, or voxel data for Static/Speed tree assets ForceVoxelData = 1, // Force generation and use of Voxel Data for click detection
VX Voxel Marker	2 bytes	Header to allow skipping.
Are Voxels Defined	1 byte (bool 0 or 1)	If this is false, skip to the next section (currently end of file).
VS Saved Voxel Marker	2 bytes	

Size of Section in Bytes	4 bytes (Integer)	
Version	4 bytes DWORD	Currently expects the value 1
Bounds Min	16 bytes (Vector 4 floats)	
Bounds Max	16 bytes (Vector 4 floats)	
Voxel Dimensions	4 bytes (Integer)	Defines N for Count of N x N x N dimensions. (e.g., 32 is typical)
Voxel Size Largest Axis	4 bytes (Float)	Scale multiplier
Count of Bytes in Voxel Data	4 bytes (Integer)	
Array of Bytes	1 byte * Count of Bytes in Voxel Data	

5.11 Auto Attach Properties

See details in section 2.2.5 for more information about these properties.

Name	Type	Info
Auto Attach Corpse To Bone	1 byte (bool 0 or 1)	Added in TMMModel version 36. Used during Sim Database generation. Not used directly during gameplay.
Corpse Attach Bone Name	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	Added in TMMModel version 36. Used during Sim Database generation. Not used directly during gameplay.
Default Death Animation Unsafe	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)1	Added in TMMModel version 36. Used during Sim Database generation. Not used directly during gameplay.

	byte (bool 0 or 1)	
Uses Auto Generated Impact Points	1 byte (bool 0 or 1)	Added in TMMModel version 36. Used during Sim Database generation. Not used directly during gameplay.
Default Idle Animation Path Unsafe	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	Added in TMMModel version 36. Used during Sim Database generation. Not used directly during gameplay.
Count of Manual Impact Points	4 bytes (Integer)	Added in TMMModel version 37. Used during Sim Database generation. Not used directly during gameplay.
Manual Impact Point Translation	CountOfManualImpactPoints occurrences of: 16 bytes (Vector 4 floats)	Added in TMMModel version 37. Used during Sim Database generation. Not used directly during gameplay.

5.12 End of TMM File

No additional data at the end of the TMM file.

6 TMM.data File Definition

The TMM.data holds all the data for the buffers defined near the start of the TMM file. The buffer offsets and counts are expected to match.

Buffers are expected to be written in the following order. However, you can put them in any order as long as the file offset in bytes and the count of bytes are set correctly.

Name
Vertex Buffer
Index Buffer
Skinning Buffer
Destruction Buffer
Color Data Buffer
Local Height Data Buffer

6.1 Vertex Buffer

Each vertex in the buffer is defined with the following data.

```

1 unsigned int posx : 16; // Low bits of 0th int
2 unsigned int posy : 16; // High bits of 0th int
3
4 unsigned int posz : 16; // Low bits of 1st int
5 unsigned int texu : 16; // High bits of 1st int
6
7 unsigned int texv : 16; // Low bits of 1st int
8 unsigned int quatx : 15;
9 unsigned int quath : 1; // Handedness
10
11 unsigned int quaty : 15;
12 unsigned int reserved0 : 1; // 1 unused bits of data
13 unsigned int quatz : 15;
14 unsigned int reserved1 : 1; // 1 unused bits of data

```

6.2 Index Buffer

Generated using the following logic:

```

1 size_t baseIndex = baseIndices[primitiveIndex];
2
3 for (size_t faceIndex : range(primitive.faces.size()))
4 {
5     for (size_t i : range(0, 3))
6     {
7         unsigned int index = primitive.faces[faceIndex].i[i];
8         indexBuffer[baseIndex + faceIndex * 3 + i] = (unsigned
9         short)index;
10    }
11 }

```

6.3 Skinning Buffer

The following definition compresses 4 bone weights and 4 bone indices into 8 bytes.

Name	Type	Info
Bone Weight 0	1 byte	Compressed float by applying $((\text{unsigned int})(\text{input.boneWeights}[0] * 255) \& 0x000000FF)$
Bone Weight 1	1 byte	Same as above.
Bone Weight 2	1 byte	Same as above.
Bone Weight 3	1 byte	Same as above.
Bone Index 0	1 byte	Compressed float by applying $((\text{unsigned int})(\text{input.boneWeights}[0] * 255) \& 0x000000FF)$ Note that position 0 includes the difference to ensure that the 4 Index values combined add up to 255.
Bone Index 1	1 byte	Same as above.
Bone Index 2	1 byte	Same as above.
Bone Index 3	1 byte	Same as above.

6.4 Destruction Buffer

Each vertex expects the following data to be present when there is any destruction data buffer. This is really a compressed unsigned short data type.

Name	Type	Info
Burnt Interior Color	6 bits	Compressed float by applying $((\text{unsigned int})(\text{value} * 255) \& 0x000000FF)$ This saves having to specify an entire color data buffer for destruction.
Destruction Bone Paired Index	10 bits	The bone index in the destroyed skeleton to pair against. Can be -1 for none.

6.5 Color Data Buffer

Each vertex expects the following data to be present when there is any color data buffer.

Name	Type	Info
R color	1 byte	Compressed float by applying $((\text{unsigned int})(\text{value} * 255) \& 0x000000FF)$
G color	1 byte	Same as above.
B color	1 byte	Same as above.
A color	1 byte	Same as above.

6.6 Local Height Data Buffer

Each vertex expects the following data to be present when there is any local height data buffer.

Name	Type	Info
Height at Vertex	2 bytes (unsigned short)	Copy of the Y value of the vertex position compressed into an unsigned short.

6.7 Speed Tree Data Buffer

Each vertex expects the following data to be present when there is any speed tree data buffer.

Name	Type	Info
Anchor X	2 bytes (unsigned short)	3D vector in local space that represents the anchor for leaves as their pivot point.

Anchor Y	2 bytes (unsigned short)	
Anchor Z	2 bytes (unsigned short)	
Geometry Type	2 bytes (unsigned short)	GeometryType_Branch = 0, GeometryType_Cap = 2, GeometryType_Leaf1 = 3, GeometryType_Leaf2 = 4, GeometryType_Mesh = 5
Wind Data X	2 bytes (unsigned short)	
Wind Data Y	2 bytes (unsigned short)	

6.8 End of TMM.data file

No additional data at end of file.

6 TMA File Definition

The following sections define a breakdown for the contents of a TMA file. TMA files represent animations that can be played on TMM models with matching skeletons.

6.1 TMA Header

Name	Type	Info
Header	4 bytes (4 chars)	Value is always "BTMA"
Version	4 bytes DWORD	Currently, version 12 is the latest and expected. Only animations matching exactly this version will currently load.
Count of Animation Tracks	4 bytes DWORD	
Resample Frame Count	4 bytes DWORD	Matches to the original frame count unless the resample frames was set to a greater than 0 value.
Animation Length in Seconds	4 bytes (Float)	Matches the original duration unless the override length modified it using a greater than 0 value.

Root Bone Bounding Box Min	16 bytes (Vector 4 floats)	
Root Bone Bounding Box Max	16 bytes (Vector 4 floats)	
Count of Bones	4 bytes DWORD	
Count of Animation Controllers	4 bytes DWORD	

6.2 Bones

The following is repeated for the “Count of Bones” quantity.

Name	Type	Info
Bone Name	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Bone Parent	4 bytes (Integer)	
Local Transform	64 bytes D3DXMATRIX	
Bind Pose	64 bytes D3DXMATRIX	
Inverse Bind Pose	64 bytes D3DXMATRIX	

6.3 Animation Tracks

The following is repeated for the “Count of Animation Tracks”.

Name	Type	Info
Track Name	String: 4 bytes DWORD size followed by length * 2	Bone name that this animation track represents.

	bytes (WCHAR)	
Version	1 BYTE	Currently expects the value 1.
Translation Encoding	1 BYTE	Constant = 0 (Only the first value is stored, and locally to avoid a pointer dereference) Raw = 1 (Stores the raw encoding of data)
Rotation Encoding	1 BYTE	Constant = 0 (Only the first value is stored, and locally to avoid a pointer dereference) Raw = 1 (Stores the raw encoding of data) Quaternion32BitCompression = 2 (Compressed 32bit quaternions) Quaternion64BitCompression = 3 (Compressed 64bit quaternions)
Scale Encoding	1 BYTE	Constant = 0 (Only the first value is stored, and locally to avoid a pointer dereference) Raw = 1 (Stores the raw encoding of data)
Count of Key Frames	4 bytes (Integer)	
Translation Data	Depends on Encoding mode.	If Encoding mode is Constant (or Count of Frames is 1): - Size is a single __m128 that holds 4 sets of 4 byte floats with X, Y, Z, and 1.0 as the last. If Encoding Mode is Raw: - Data is stored as “Count of Key Frames” multiplied sizeof(D3DXVECTOR3) quantity of unsigned chars that hold sequential X, Y, Z values. D3DXVECTOR3 is 12 bytes.
Rotation Data	Depends on Encoding mode.	If Encoding mode is Constant (or Count of Frames is 1): - Size is a single __m128 that holds 4 sets of 4 byte floats with X, Y, Z, W . If Encoding Mode is Raw: - Data is stored as “Count of Key Frames” multiplied sizeof(D3DXQUATERNION) quantity of unsigned chars that hold sequential X, Y, Z, W values. D3DXQUATERNION is 16 bytes. If Encoding Mode is Quaternion32BitCompression: - Data is stored as “Count of Key Frames” * sizeof(uint32_t). Where each Quaternion is compressed to fit into a uint32. (4 bytes) If Encoding Mode is Quaternion64BitCompression: - Data is stored as “Count of Key Frames” * sizeof(uint64_t). Where each Quaternion is compressed to fit into a uint32. (8 bytes)
Scale Data	Depends on Encoding	If Encoding mode is Constant (or Count of Frames is 1):

	mode.	- Size is a single <code>__m128</code> that holds 4 sets of 4 byte floats with X, Y, Z, and 1.0 as the last. If Encoding Mode is Raw: - Data is stored as “Count of Key Frames” multiplied <code>sizeof(D3DXVECTOR3)</code> quantity of unsigned chars that hold sequential X, Y, Z values. <code>D3DXVECTOR3</code> is 12 bytes.
--	-------	--

6.4 Animation Controllers

The following is repeated for the “Count of Animation Controllers”. The data format varies depending on what type the controller is. The first value is the type and that tells you which table to use.

6.4.1 Visibility Animation Controller

Name	Type	Info
Type	4 bytes (Integer)	Value = 1
Start Time	4 bytes (Float)	
End Time	4 bytes (Float)	
Ease In Time	4 bytes (Float)	
Ease Out Time	4 bytes (Float)	
Invert Logic	1 byte (bool 0 or 1)	
Attach Point Name	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	

6.4.2 Footprint Animation Controller

Name	Type	Info
Type	4 bytes (Integer)	Value = 2
Spawn Time	4 bytes (Float)	
Footprint Name	String: 4 bytes DWORD size followed by length * 2	

	bytes (WCHAR)	
Footprint ID	4 bytes (Integer)	
Invert Texture Y	1 byte (bool 0 or 1)	
Attach Point Name	String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	
Is Right Side	1 byte (bool 0 or 1)	

6.5 Errors

The following defines if there are any errors that were logged during processing of output.

Name	Type	Info
Error Flags	4 bytes DWORD	<pre> 1 Default = 0, (No errors) 2 AnimationMismatchWithSourceSkeleton = (1 << 0), 3 AnimationHadNaNInFrames = (1 << 1), </pre>
Count of Error Details	4 bytes DWORD	
Error Detail Strings	Repeated: String: 4 bytes DWORD size followed by length * 2 bytes (WCHAR)	For each “Count of Error Details”: String defining information about the error.

6.6 End of TMA file definition.

No additional data at end of file.

7 DEPS File Definition

The deps file defines dependency relationship between files. It has no functional purpose and in development environments is used for determining if a model needs rebuilding during scanning for changed files. Each of the following appear as one or more lines in the file.

Name	Description
------	-------------

Version	Expects: V, 3 This is just saying that the version is “3”.
Targets	Any file that is an output result of the export process. Expects form: T, %s, %lld Where “T” refers to Target, %s is the path to that target, %lld is the modified time of the file.
Dependencies	Any file that is an input of the export process. Expects form: D, %s, %lld Where “D” refers to Dependency, %s is the path to that dependency, %lld is the modified time of the file.